

Unified Consciousness Theory I

Ontic Structure, Physical Views, and the Structural Continuity from Physical Process to Conceptual Self

Author: Hongju Liu

Affiliation: Independent researcher, Shenzhen, China

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Versioning statement. This manuscript is a revised successor to *Unified Consciousness Theory I: From Experience Existence to Experiential Structure*, v1.0. It does not overwrite or retroactively alter v1.0. The main purposes of v1.1 are to (i) separate ontic process structure from scientific views and estimates, (ii) state explicitly the strength of the structural-identity commitment used by later derivations, (iii) repair theorem dependencies, (iv) formalize the distinction between metaphysical identity and finite empirical bridge tests, (v) develop a structural-evolutionary synthesis from elementary physical processes to human conceptual selfhood, and (vi) align UCT II through a typed scientific-view interface and an exact structural-retention example. Evolution is not treated as a proof of the foundational axioms. Internal candidate labels rc3 and rc4 refer to the preserved development history, not additional published editions.

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Abstract

Unified Consciousness Theory (UCT) adopts nonempty experience for every actual valid physical process token and treats that experience as the intrinsic presentation of the process's own organization. A cell and a human are interpreted by the same principle. Evolutionary change concerns the relations, discriminations, temporal organization, and self-representations instantiated by processes, rather than the arrival of a new experiential substance at a privileged biological stage.

This revision makes the intended structure-for-structure identity explicit. For each actual process P , its complete physical and experiential organizations, expressed in a declared common structural signature $\mathcal{K}$, are isomorphic:

$$\Phi_{\mathcal{K}}(P) \cong \mathbf{D}_{\mathcal{K}}^{\text{ontic}}(P).$$

The statement, denoted A5-SI, formalizes one organization under physical description and intrinsic experiential presentation. It is stronger than the pairwise type-equivalence formula displayed in rc3; that weaker formula is now a consequence. A finite countermodel explains why the distinction matters. Under A5-SI, standard products of constituent isomorphisms yield a corrected non-summative combination theorem without adding an independent experience-generating mechanism. Non-product organization concerns a specified constituent comparison, not a numerical subject count.

Actual process plurality is accepted at all relevant physical scales. Macroprocess analysis describes realized causal organization; it does not introduce another experience-existence gate. A unique universal partition into subjects is not a prerequisite of the theory. A scoped symmetry-orbit result constrains optional exclusive subject selectors while retaining possible invariant global partitions. Human conceptual selfhood is treated as higher-order organization within experience.

A three-register thought experiment has identical declared input-output behavior in two circuit settings while its complete modeled transition maps are nonisomorphic. An implemented sub-process retains its organization. This exactly checked case distinguishes whole-process change from sub-process preservation without changing the comparison token after an outcome. An extension with separately controlled cross-links proves three exact internal-retention regimes: immediate erasure, one-step retention, and sustained distinguishability, despite unchanged declared output. The mathematical models supply no independent phenomenal measurements.

The framework separates ontic organization from finite scientific views and estimates. For later empirical applications, physical sufficiency for a measured target and reflection of all relevant structural differences are distinct bridge requirements. The remaining research program is to derive more specific structural consequences and faithful comparisons with neighboring theories. It does not require a second account of why human rather than cellular experience exists. The paper presents a unified explanatory principle with conditional formal consequences; empirical confirmation and complete intertheory recovery are separate claims.

Keywords: consciousness; panexperientialism; process ontology; structural identity; composition; evolution; experiential structure; self-model; scientific views

1. The target problem

Contemporary consciousness theories assign central roles to integrated causal structure, recurrent processing, global availability, higher-order representation, predictive modeling, information closure, temporal organization, and related mechanisms (Seth & Bayne, 2022; Albantakis et al., 2023; Chang et al., 2020; Kanai & Fujisawa, 2024). Much of the competition among theories concerns whether one of these mechanisms constitutes, enables, or marks consciousness.

UCT begins by separating two questions:

Does experience exist?

and:

What is the organization of that experience?

UCT fixes the first question axiomatically on its declared domain of actual process tokens. It therefore relocates the nontrivial universal problem to experiential structure.

Universal experience and structural identity are the adopted explanatory starting points. Internal derivations are assessed by whether they follow from those commitments. The substantive comparative task is to constrain structured differences, multiscale relations, and transformations; empirical applications additionally require declared bridge predictions. Human and cellular experience do not require different existence mechanisms within this framework.

The principal clarification of v1.1 is that the identity claim concerns the actual physical organization of a process token, not a representation chosen by an analyst. The second is that finite experiments do not directly observe complete ontic organization or complete phenomenology. Operational tests therefore target explicitly declared finite bridge hypotheses.

2. Ontological domain: actual process tokens

2.1 Actuality and tokenhood

Let:

$$\mathcal{P}$$

be the domain of actual valid physical process tokens. An element:

$$P \in \mathcal{P}$$

is an actual causally continuous physical subhistory, not an arbitrary set of variables selected after the fact.

A valid token must satisfy, at minimum:

1. **actual support** — the relevant states/events are physically instantiated;
2. **induced-relation fidelity** — attributed internal relations are inherited from physical organization rather than invented by the modeller;
3. **causal continuity** — the token has a physically traceable causal history over the interval of interest;
4. **boundary accountability** — materially relevant cross-boundary influences are represented rather than silently erased;
5. **lineage traceability** — continuation, replacement, branching, merger, and cessation are physically trackable;
6. **description invariance** — invertible coordinate or label changes do not create new tokens.

UCT is process-token pluralist. Actual tokens may:

- overlap;
- nest;
- branch;
- merge;
- persist through replacement;
- exist at multiple physical scales.

No maximality or exclusion rule is imposed at the basal level.

2.2 Multiscale token plurality

A critical distinction in v1.1 is:

actual macroprocess token $\neq$ arbitrary coarse description.

Molecular, cellular, circuit-level, neural-system, organism-level, and other macroprocesses may all be actual process tokens when the relevant organization is physically instantiated. A scientist does not create a macroprocess merely by averaging microscopic variables.

Macroprocess is a scale-relative description of actual organized physical activity. Its experience does not wait for a new threshold of autonomy, recurrence, closure, complexity, or subjecthood. Many overlapping and nested processes may qualify; UCT does not require a unique privileged macro boundary. In an application, one must specify which actual causal organization is being described, just as one must specify which physical process a model concerns. That identification task should not be confused with a missing mechanism that turns macroprocesses into experience.

2.3 Token, type, and lineage

UCT distinguishes:

structural type $\neq$ numerical token $\neq$ causal lineage.

Two process tokens can have isomorphic organization without being numerically identical:

$$\mathbf{D}(P) \cong \mathbf{D}(Q) \not\Rightarrow P = Q.$$

Lineage is a directed continuation relation rather than numerical identity. A process may branch:

$$P \rightsquigarrow Q_1, \quad P \rightsquigarrow Q_2, \quad Q_1 \neq Q_2,$$

or contribute to a merger:

$$P \rightsquigarrow M, \quad Q \rightsquigarrow M.$$

Nothing in these relations alone determines a unique subject count.

2.4 A concrete finite process criterion

In a finite classical description, let G represent physically instantiated events and their causal relations. An admissible process subhistory H consists of actual events, inherits their physical relations, is connected through the inherited causal links over the declared interval, and records materially relevant links across its boundary. Connectivity here does not require mutual influence between every pair, maximality, an autonomous Markov state, or a nervous system. A numerical renaming of H does not create another physical occurrence.

This is a concrete expression of Section 2.1's process domain, not an independent consciousness detector. Whenever such a process is actual, A1 applies. Different subhistories can overlap and nest; selecting all admissible processes does not demand an exclusive partition. Greater integration, temporal depth, differentiation, or self-maintenance then characterizes differences of organization within the domain.

The representation G must itself be grounded in the physical system. A convenient calculated feature may describe an existing process without creating an additional token. In contrast, an actual causal relation remains relevant even when it crosses an observer's preferred anatomical or computational boundary. Applications may investigate a process's current state carriers, dynamical contributors, enabling conditions, and output recipients separately. Those roles do not determine whether some underlying actual local process has experience.

A cell, a multicellular organism, and an extended interaction can therefore each instantiate experience-bearing organization without passing three different ontological tests. An entire organism need not be one indivisible subject for its many realized processes to fall within UCT. Appendix D gives optional procedures for checking finite coarse descriptions; their predictive-closure conditions are not required by the process criterion or by A1.

3. Ontic organization, views, estimates, and subjects

The central formal firewall of v1.1 is:

$$\text{ontic process token} \neq \text{physical view} \neq \text{scientific estimate} \neq \text{subject}.$$

3.1 Full token-relative ontic organization

For an actual process token P , define:

$$\mathbf{D}_t^{\text{ontic}}(P)$$

as the complete currently instantiated physical organization constitutive of that token.

"Complete" is token-relative. It means that every physical distinction or relation constitutive of P 's own organization is included. It does not mean that every microscopic fact anywhere in the realizing substrate or surrounding universe is part of the same token.

This distinction permits fine and coarse actual process tokens to coexist without making every microscopic fluctuation a mandatory difference in a human-scale macroexperience.

3.2 Physical scientific views

Science does not ordinarily access $\mathbf{D}^{\text{ontic}}(P)$ directly. Instead, it constructs physically anchored views indexed by:

$$v = (t, \tau, g, \mathcal{I}, \mathcal{B}, \dots),$$

where t is time, τ a horizon, g a measurement/representation grain, $\mathcal{I}$ a physically admissible intervention family, and $\mathcal{B}$ a declared boundary representation.

Write:

$$D_v(P) = \Pi_v[\mathbf{D}^{\text{ontic}}(P)].$$

Different views need not form one total hierarchy. Some are refinements or coarse projections of others; some are incomparable.

Where two routes claim to construct the same target physical view, local consistency requires approximate or exact commutation:

$$R_{v \rightarrow z} \cong R_{w \rightarrow z} \circ R_{v \rightarrow w}.$$

3.3 Scientific estimates

Actual data yield:

$$\widehat{D}_v(P \mid \mathcal{E}, \mathcal{M}),$$

an estimate conditioned on evidence $\mathcal{E}$ and model class $\mathcal{M}$.

If evidence underdetermines the physical organization, the appropriate scientific output is an identified set:

$$\mathfrak{I}(P) = \{D : D \text{ remains compatible with the available evidence}\},$$

rather than an ontological point estimate chosen for convenience.

3.4 Subjecthood

A subject is not defined merely by being a process token, a scientific view, or a macro-process.

UCT therefore maintains:

experience-bearing process $\neq$ canonically individuated subject.

A universal positive subject selector is not required for this basal theory. Where a study uses subject to mean an experience-bearing process, the process domain supplies its referent. Where it means a human-like owner or conceptual self, the relevant self-organizing and self-representational relations belong to higher-order organization. Neither usage licenses imposing one exclusive subject partition on all actual processes.

3.5 A pointed, port-aware comparison signature

The word *structure* must not conceal what an isomorphism is required to preserve. In the finite controlled models below, the declared signature has sorts for states, boundary inputs, intervention ports, outputs, constituent slots, and time; operation symbols for update, reset, readout, and constituent projection; a distinguished current-state constant; and the declared temporal and incidence relations. A model in that signature can be written

$$\mathcal{M}_v = (X, x_\bullet, U, J, Y, T, I, o, \mathcal{C}, \preceq).$$

The tuple is a model **in** the signature, not the signature itself. Here $T_u : X \rightarrow X$ is an update at fixed constitutive mechanism, $I_j : X \rightarrow X$ a permitted state reset, and $o : X \rightarrow Y$ a declared output. $\mathcal{C}$ specifies constituent projections and physically identified ports. Current-state pointing is essential: two occurrences of the

same transition law at inequivalent current states need not have the same complete organization.

An admissible isomorphism contains bijections f_X, f_U, f_J, f_Y that satisfy

$$f_X(x_\bullet) = x'_\bullet, \quad f_X T_u = T'_{f_U(u)} f_X, \quad f_X I_j = I'_{f_J(j)} f_X, \quad f_Y o = o' f_X,$$

and preserve the declared temporal relations and constituent incidence. A port permutation is allowed only when it is declared as a symmetry of the comparison; an arbitrary mixture of physically distinct ports is not a relabeling of a fixed constituent split. Changing a mechanism, removing an intervention, or reversing a directed temporal relation generally changes the comparison rather than supplying an isomorphism. A probe family chosen by the analyst remains part of a scientific view, whereas a process's actual causal dispositions are ontic.

For the product comparison in Section 6, hold the common boundary convention fixed, use the product state space with its pointed state, and retain the specified component-local update and reset operations. These declarations identify the ordinary independent product used on both sides of A5-SI. They do not impose autonomy, reciprocity, maximality, or a new threshold for experience.

The finite signature supplies an explicit example, not a claim that a finite model captures all constitutive properties of real hardware or a brain. A restricted signature or measured view can omit distinctions. Completeness relative to an actual token is therefore not inferred from success in this finite representation. In biological applications the relevant state carriers, interfaces, and time relations must be justified independently; the same foundational identity principle is used.

4. Foundational commitments

A1 — Universal Experience

For every actual valid process token:

$$\boxed{\forall P \in \mathcal{P}, \quad \Phi(P) \neq \emptyset.}$$

A1 is not derived from evolution, neuroscience, intelligence, report, complexity, or biology.

The weaker equivalence formulas A5-W and A5-OI do not themselves establish nonemptiness for every actual token. Under a strong structure-for-structure representation, nonemptiness may also follow mathematically from nonempty physical structure. A1 is retained to state the intended ontological commitment and to prevent actual-process definitions from quietly excluding simple cases.

A2 — Scoped Structural Continuity

For an explicitly declared continuously comparable family $P(\lambda)$, if the relevant physical organization varies continuously in a specified structural topology, then experiential organization varies continuously:

$$\boxed{\mathbf{D}(P_\lambda) \text{ continuous} \Rightarrow \Phi(P_\lambda) \text{ continuous.}}$$

A2 is a schema, not a universal scalar continuity law. Quantitative use requires a declared metric/topology. A derived categorical observable can undergo a threshold transition even when underlying structure varies continuously.

A3 — Persistence Under Embedding

If an actual process P remains physically realized while embedded in a larger actual process Q , embedding alone does not terminate P :

$$P \triangleleft Q \not\Rightarrow \text{cessation of } P.$$

The distinctive content of A3 is process ontological. Once the lower token remains actual, A1 supplies nonempty lower-scale experience.

A3 does not imply that a process continues after its constitutive dynamics cease.

A4 — Selfhood Non-Prerequisite

Human-like conceptual selfhood is not required for basal experience:

$$\neg \text{Self}_I(P) \not\Rightarrow \Phi(P) = \emptyset.$$

The positive mechanisms by which body models, agency, autobiographical memory, metacognition, and a conceptual “I” form are not axioms. They belong to higher-order application/bridge layers.

A5 - Experience-Structure Identity

UCT’s intended claim is that experience is the intrinsic presentation of an actual physical process’s own organization. Cellular and human experience instantiate this same identity principle; richer organization does not require a second kind of experiential ingredient.

A5-SI - explicit structure-for-structure identity

Fix a common structural signature $\mathcal{K}$ for the comparison: the relevant state distinctions, causal and temporal relations, and constituent organization must be specified. Write $\mathbf{D}_{\mathcal{K}}^{\text{ontic}}(P)$ and $\Phi_{\mathcal{K}}(P)$ for the complete physical and experiential organization of the same actual token in this signature. The strong intended identity is expressed by

$$h_P : \mathbf{D}_{\mathcal{K}}^{\text{ontic}}(P) \cong \Phi_{\mathcal{K}}(P).$$

The two expressions describe the same organization in physical and experiential terms. The isomorphism is a formal expression of that commitment, not an empirically fitted correspondence, an extra causal force, or an experience-generating mechanism. Structural relations must be fixed by the declared signature; they cannot be freely relabeled to fit a desired phenomenal result. When an independent product is used, it is the ordinary product in the common comparison category on both sides. Section 6 makes this typing explicit.

Version boundary. This is stronger than the pairwise equivalence formula displayed in rc3. The rc4 development revision adopted it to formalize the intended identity

interpretation; it does not claim to have deduced it from the weaker formula or to have established it empirically. The published v1.0 and the original rc3 remain unchanged.

A5-OI - pairwise type equivalence as a consequence

Within that common signature, composing the tokenwise isomorphisms gives

$$\Phi(P) \cong \Phi(Q) \iff \mathbf{D}^{ontic}(P) \cong \mathbf{D}^{ontic}(Q).$$

The subscripts are suppressed when the scope is clear. This consequence preserves and reflects equivalence between actual token types, but if written alone it is weaker than A5-SI: an arbitrary permutation of type labels can satisfy it without preserving each token's organization. Section 6 supplies an explicit counterexample to that weaker reading.

A5-W - one-way phenomenal completeness

The one-way consequence inherited from the explicitly stated v1.0 formula is

$$\mathbf{D}(P) \cong \mathbf{D}(Q) \implies \Phi(P) \cong \Phi(Q).$$

It excludes a freely varying hidden phenomenal variable when complete constitutive physical organization is fixed. No product conclusion is attributed to this one-way formula alone. In the rest of the paper, A5 denotes the intended strong identity, with A5-W or A5-OI cited when a weaker consequence already suffices.

A6 — Token / Type / Lineage Distinction

$$\boxed{\mathbf{D}(P) \cong \mathbf{D}(Q) \not\Rightarrow P = Q.}$$

Structural equivalence does not imply numerical identity.

5. Formal consequences

5.1 Organization-Gate Impossibility

Define an experience-existence predicate:

$$E(P) = \mathbf{1}[\Phi(P) \neq \emptyset].$$

By A1:

$$E(P) = 1 \quad \forall P \in \mathcal{P}.$$

Let $F(P)$ be any organization descriptor and S a threshold/target region. If:

$$\exists Q \in \mathcal{P} : F(Q) \notin S,$$

then:

$$\boxed{\mathbf{1}[F(P) \in S] \neq E(P).}$$

Hence, within UCT, no nonuniversal threshold of:

- integration;
- recurrence;
- global availability;
- explicit memory;
- higher-order representation;
- predictive modeling;
- autonomy;
- entropy;
- generic biological complexity;

can equal the universal experience-existence predicate.

This theorem does not make those mechanisms irrelevant. It relocates them from experience existence to experience organization.

5.2 No Hidden Quale

From A5-W:

$$\mathbf{D}^{ontic}(P) \cong \mathbf{D}^{ontic}(Q) \Rightarrow \Phi(P) \cong \Phi(Q).$$

UCT therefore rejects an additional hidden phenomenal degree of freedom that varies independently while complete token-relative physical organization is fixed.

Within the intended structuralist interpretation, words such as “red” and “pain” label experiential organizations rather than additional freely varying quale atoms. A5-W alone does not determine the physical organization corresponding to either label or establish the complete structuralist interpretation.

5.3 Probe Invariance

Let $\mathcal{J}^{exp}$ denote an experimenter-selected probe family. Merely changing this choice, without changing the complete constitutive physical organization, cannot alter phenomenology under A5-W:

$$\begin{aligned} \mathcal{J}_1^{exp} &\neq \mathcal{J}_2^{exp}, \\ \mathbf{D}_1^{ontic}(P) \cong \mathbf{D}_2^{ontic}(P) &\implies \Phi_1(P) \cong \Phi_2(P). \end{aligned}$$

The completeness qualifier matters. Holding only one causal graph or one response descriptor fixed is insufficient if other constitutive state distinctions change. A probe that physically intervenes on the process can change its experience by changing its organization; the invariant claim concerns a merely external change of description or choice.

5.4 No Future Contamination

If:

$$\mathbf{D}_t^{ontic}(P) \cong \mathbf{D}_t^{ontic}(Q),$$

then a future external divergence that is not physically instantiated at time t cannot imply a present phenomenal difference:

$\Phi_t(P) \cong \Phi_t(Q).$

Present causal dispositions are part of the present organization; actual future facts are not.

5.5 Causally Screened History / No Ghost History

Suppose:

$$History(P) \neq History(Q)$$

but all current constitutive physical consequences of those histories have converged:

$$\mathbf{D}_t^{ontic}(P) \cong \mathbf{D}_t^{ontic}(Q).$$

Then:

$$\boxed{\Phi_t(P) \cong \Phi_t(Q)}.$$

A6 still permits the tokens to remain numerically distinct because lineage can differ even when current structural type is the same.

5.6 External reward-label non-identity

Changing an external reward sign, semantic label, or experimenter convention without changing intrinsic constitutive organization cannot by itself alter phenomenology under A5-W.

Therefore:

$$\boxed{\text{reward sign} \neq \text{phenomenal valence sign.}}$$

6. Factorization, macroprocesses, and combination

6.1 Independent products and comparison objects

Fix structural signatures and a constituent split before comparing a whole with its parts. The physical product $X_1 \otimes \cdots \otimes X_n$ represents independent juxtaposition relative to that split and a declared boundary convention. In a transition model, component updates must not depend on the other component states once the declared boundary is fixed. A shared drive can create statistical correlations without creating the internal causal dependence at issue. Product factorization is consequently not synonymous with unconditional statistical independence.

For this product comparison, constituents occupy nonoverlapping slots of the specified physical split. A product of overlapping actual tokens would require a separately justified comparison construction; general gluing of overlapping processes is not established here. An abstract product is initially a **comparison object**. It need not itself be an actual causally continuous token; A1 is not automatically applied to it. Unrestricted mixing of physically distinct intervention ports must not be mistaken for a mere relabeling that preserves a particular constituent split. The phenomenal product must also be specified as the intended independent juxtaposition, not retrospectively defined to make a desired conclusion true.

6.2 Why A5-OI alone is insufficient

A5-OI constrains equivalence of structural types. It does not state how products are transported. The distinction is visible in a finite-set countermodel. Take isomorphism classes indexed by positive cardinality and the bijection

$$f(6) = 7, \quad f(7) = 6, \quad f(n) = n \quad (n \neq 6, 7).$$

Let physical constituent types have sizes 2 and 3, and a whole type have size 7. Map phenomenal types using f . The equivalence biconditional holds because f is injective. Nevertheless,

$$7 \neq 2 \cdot 3, \quad f(7) = 6 = f(2) \cdot f(3).$$

Thus a nonproduct physical type can correspond to the product of the constituent phenomenal types while the bare A5-OI equivalence formula is satisfied. This is an abstract countermodel to an implication of the formula, not a claim that unstructured finite sets are adequate physical or phenomenal models. It pinpoints the missing algebraic commitment. Additional physical restrictions may exclude it, but those restrictions must be stated.

6.3 Non-summative combination from the intended identity

Use the strong tokenwise identity A5-SI in a common comparison category with the finite products needed for the argument. Independent juxtaposition on each side is represented by this same standard product. Ordinary products preserve isomorphisms: if each $h_i : X_i \cong Y_i$, then

$$\bigotimes_i h_i : \bigotimes_i X_i \cong \bigotimes_i Y_i.$$

This is the relevant mathematical consequence of expressing the same constituent organizations in two descriptions. It is not another physical mechanism of experiential combination. The use of a common signature and product must be declared; the weak type-equivalence formula alone cannot supply it.

Theorem 6.3 (non-summative combination in the declared product comparison). Let M be an actual whole process and P_i its actual constituents in the nonoverlapping slots of the specified comparison. If

$$\mathbf{D}^{ontic}(M) \not\cong \bigotimes_i \mathbf{D}^{ontic}(P_i),$$

then A5-SI implies

$$\Phi(M) \not\cong \bigotimes_i \Phi(P_i).$$

Proof. Let h_M and h_i be the tokenwise isomorphisms supplied by A5-SI. If $\Phi(M) \cong \bigotimes_i \Phi(P_i)$, compose that isomorphism with h_M and the inverse of the product of the h_i . This yields

$$\mathbf{D}^{ontic}(M) \cong \Phi(M) \cong \bigotimes_i \Phi(P_i) \cong \bigotimes_i \mathbf{D}^{ontic}(P_i),$$

contradicting the physical premise. The abstract comparison product need not be an actual token, so no extension of A1 to imaginary processes is used. $\square$

The result says that actual additional constitutive relations are not erased in the experiential description. It does not add together subjective substances. Its nonproduct conclusion is relative to the specified constituent product, not every possible decomposition. Continuing constituent processes remain experience-bearing by A1 and A3. Neither the theorem nor the theory requires a unique numerical subject count before admitting the whole's experiential organization.

If identity is deliberately weakened back to the pairwise A5-OI formula alone, the countermodel in Section 6.2 still applies and this proof is unavailable. The repair therefore makes the intended premise explicit rather than silently inferring a stronger premise from a weaker formula.

6.4 Macro invariance with micro change

Suppose actual macroprocess tokens obey

$$\mathbf{D}^{ontic}(M_A) \cong \mathbf{D}^{ontic}(M_B),$$

while an independently identified pair of lower-scale actual tokens obeys

$$\mathbf{D}^{ontic}(p^A) \not\cong \mathbf{D}^{ontic}(p^B).$$

A5-W suffices for $\Phi(M_A) \cong \Phi(M_B)$. The converse component of A5-OI additionally yields $\Phi(p^A) \not\cong \Phi(p^B)$. Only the stated pairwise consequences are needed for these claims; the product construction is not involved.

The result is conditional: it does not show that any specified material replacement actually preserves complete macro organization. Nor can one establish it merely by observing the same coarse statistic of the same token. The macro token must have an independently supported realization and boundary; lower-scale physical differences must not be constitutive differences of that macro token. Appendix B demonstrates the corresponding distinction between a fixed whole-process token and an invariant implemented sub-process in an idealized finite model. It is not a general biological replacement theorem.

7. Subject selection under symmetry: an orbit obstruction

UCT distinguishes experience, structural unity, macroprocess tokenhood, subjecthood, and self-modeling. The result below concerns a proposed selection rule. It neither derives subjecthood from A1 nor treats every experience-bearing process as one canonical subject.

7.1 Equivariant selection is a union of candidate orbits

Let W be a physical structure and Γ a group of its structure-preserving automorphisms. Let $\mathcal{C}(W)$ be a candidate family closed under this group. A deterministic selector returns $S(W) \subseteq \mathcal{C}(W)$ and is equivariant:

$$S(gW) = gS(W).$$

Since $gW = W$ for $g \in \Gamma$, the selected family is invariant under Γ .

Orbit lemma. If a candidate P is selected, its entire orbit ΓP is selected.

Proof. For each $g \in \Gamma$, invariance gives $gP \in gS(W) = S(W)$. Thus a selected family is a union of full orbits. $\square$

Corollary 7.1 (restricted exclusive-selection obstruction). Suppose the admissible candidates are one orbit O containing an overlapping pair. No deterministic equivariant selector can return a nonempty, pairwise disjoint subfamily of O .

Proof. A nonempty selection contains one candidate and hence all of O . It then contains the overlapping pair, contrary to pairwise disjointness. $\square$

This corollary requires the stated restriction of the candidate family and a nonempty-output requirement. These are selection requirements, not consequences of A1-A6.

7.2 The three-part example and its counterlimits

For a reflection-symmetric world with parts a, b, c , let the reflection exchange a and c , and let the candidate orbit be

$$O = \{\{a, b\}, \{b, c\}\}.$$

The two invariant selected subfamilies are the empty family and the whole orbit. The first violates nonemptiness; the second violates exclusivity. An unbiased random rule could choose either member with equal probability, but it would not determine one unique subject from W ; a realized random seed would add selection data.

The theorem does **not** rule out the invariant partitions

$$\{\{a, b, c\}\}, \quad \{\{a\}, \{b\}, \{c\}\}.$$

If a positive subject theory admits one of these, it needs its own physical justification. Neither is excluded by the orbit lemma. The correct conclusion is a scoped obstruction to selecting nonempty exclusive candidates from an overlapping symmetry orbit, not a general theorem that intrinsic structure cannot determine any unique partition. A universal positive subject criterion is not imposed by UCT. Any additional exclusive selector must justify its own scope and purpose.

8. From physical process to conceptual self: a structural-evolutionary synthesis

This section is explanatory rather than foundational.

It asks:

If A1 and the intended identity A5-SI are granted, can one describe a continuous and branching path from elementary physical processes to human conceptual selfhood without introducing successive new phenomenal substances?

UCT can represent such a path coherently as a structural synthesis. This does not establish that the historical transitions occurred in exactly the sequence listed below, nor does it provide independent evidence for A1 or A5-SI. The relevant theoretical claim is narrower: no additional phenomenal substance is required by the UCT formalism when these organizational transformations are described.

Evolution supplies the organizing rationale for continuity across these cases. UCT takes universal experience and structural identity as its starting commitments; it does not repeatedly reopen their derivation inside each biological application. Comparative evolutionary findings illustrate changes of organization. Accepting biological evolution by itself need not force another theorist to adopt UCT's axioms; that external disagreement is distinct from whether UCT's internal explanations consistently use them.

8.1 Elementary physical processes

At the most elementary level considered by UCT, an actual interaction process possesses whatever relational structure is physically instantiated.

UCT does not require:

- a nervous system;
- a sensor;
- a memory;
- a self-model;
- a report channel.

Under A1, the experience-existence predicate is already nonzero. A5-SI identifies this process's experiential organization with its actual organization. The pairwise A5-OI formula is a consequence of that stronger statement.

UCT therefore refuses to project human terms such as "seeing," "wanting," or "suffering" onto elementary interactions. The primitive claim is structural, not anthropomorphic.

8.2 From stable chemistry to self-maintaining organization

The transition from nonliving chemistry to the first cells is an origin-of-life problem, not a solved Darwinian lineage. UCT does not claim to reconstruct the historical path.

The theoretical point is narrower. If physical processes develop persistent boundaries, recurrent chemical organization, and endogenous regulation, then new relations appear inside the process organization:

internal state $\leftrightarrow$ boundary $\leftrightarrow$ environmental perturbation $\leftrightarrow$ regulatory response.

These relations are not the origin of experience under A1. They are transformations of experiential organization.

8.3 Single-cell and non-neural learning: temporal depth

A decisive organizational transition occurs when past events leave present physical traces that modify current response:

$$History \rightarrow m_t \rightarrow Response_t.$$

This is the minimal form of temporal depth required by UCT's memory analysis.

The framework does not require memory to create experience. Memory changes current organization.

Non-neural learning illustrates the physical possibility of history-dependent behavior without a brain. Habituation has been experimentally demonstrated in the slime mould *Physarum polycephalum* (Boisseau, Vogel, & Dussutour, 2016). Such evidence does not establish phenomenology in the organism; it demonstrates that learning-like temporal organization need not await centralized nervous systems.

If all physical consequences of a history are erased, No Ghost History predicts no remaining current phenomenal-type difference attributable solely to the past.

8.4 Multicellularity: from juxtaposition to organism-scale relations

Consider independent cells:

$$C_1, \dots, C_n.$$

If their joint organization remains a product, there is no new irreducible cross-cell relation merely because the cells are spatially adjacent.

If, however, real chemical, electrical, mechanical, developmental, or other causal relations generate an actual multicellular process M with:

$$\mathbf{D}(M) \not\cong \bigotimes_i \mathbf{D}(C_i),$$

then the structure-for-structure identity A5-SI assigns M an experiential organization that is nonproduct relative to that constituent comparison, by Theorem 6.3. The multicellular case uses the same identity principle as the cellular case; no new experiential substance or threshold is introduced.

The significance of multicellularity in UCT is therefore not cell count. It is the formation of new actual macroprocess organization.

No privileged signaling medium is required. Mechanical, chemical, electrical, and neural relations can all contribute when they are physically constitutive.

8.5 Nervous systems: routing and specialization, not an experience switch

The evolution of neurons adds high-speed, long-range, selectively routed, and plastic coordination.

Under UCT:

$\text{neuron} \neq \text{experience ignition.}$

Neurons transform organization. They can increase spatial discrimination, routing, temporal coordination, learning capacity, and context sensitivity.

Associative learning in the box jellyfish *Tripedalia cystophora*, a cnidarian without a vertebrate-style centralized brain, provides an empirical illustration that nontrivial learning need not wait for vertebrate-like centralization (Bielecki et al., 2023).

This finding is not evidence for UCT's A1. It illustrates the danger of equating a particular level of centralized neural architecture with the first appearance of complex learning.

8.6 Damage regulation, nociception, and affect

The pain lineage is particularly useful because it exposes the mistake of projecting a fully human quale backward.

Comparative biology indicates that damage detection and adaptive cellular stress responses predate nervous systems, while specialized nociceptors, sensitization, central aversive states, and human pain involve progressively richer mechanisms (Walters, 2025).

A schematic structural trajectory proceeds through:

1. cellular damage regulation;
2. multicellular injury coordination;
3. specialized nociception and sensitization;
4. aversive learning and multidimensional affective pain;
5. human suffering.

UCT does not identify the first term with human pain.

It instead asks which physical relations constituting later pain are already present in simpler systems.

Human pain also illustrates why one scalar valence variable is inadequate. Clinical and conceptual literatures distinguish sensory-discriminative from affective-motivational dimensions of pain, and pain-asymbolia cases have been interpreted as evidence for partial dissociation between them (Auvray, Myin, & Spence, 2010). That interpretation is not uncontested: recent philosophical reviews emphasize that asymbolia may reflect strongly reduced rather than literally absent negative affect, and that the taxonomy of the residual state remains debated (Coninx, 2024). The conservative UCT conclusion is therefore:

Nociceptive discrimination, unpleasantness, and suffering should not be collapsed into one primitive scalar variable.

Under A5-SI, unpleasantness belongs to the intrinsic organization of the relevant actual process, just as any cellular quality does. Identifying which concrete human mechanisms instantiate a specified pain experience is an application-level biological question; it does not require another foundational account of why human experience exists.

One explanatory principle across experiential qualities

Let P_c be a cellular process and P_h a human process. UCT applies the same relation $\Phi_{\mathcal{K}}(P) \cong \mathbf{D}_{\mathcal{K}}^{ontic}(P)$ in each appropriately declared comparison. The organizations differ in their realized distinctions and relations; the explanatory principle does not change. Human red, pain, and bodily ownership can occupy richly differentiated positions within a process's experiential organization without requiring that a cell possess those same human discriminations or linguistic labels.

In a continuously comparable transformation family, A2 supplies experiential continuity and A5 supplies structural identity. Added, removed, or reorganized causal relations correspond to changes in the intrinsic organization itself. The theory need not posit a point where an otherwise nonexperiential organization suddenly acquires a new phenomenal substance. Nor does it follow that evolution must monotonically increase one scalar amount of experience.

At this explanatory level, asking for a second mechanism connecting a fully specified actual organization to its experience would reopen what A5 fixes as the identity principle. A separate question is which actual organization realizes a particular familiar human experience. Investigating that empirical instance can refine an application without constituting a missing axiom of the general theory.

8.7 Association, relational abstraction, and reasoning

Cognitive evolution can likewise be represented as transformations of organization rather than the sudden arrival of a new phenomenal substance.

A simplified comparative sequence runs from state-dependent response through habituation and sensitization, association, relational abstraction, relations over relations, and symbolic or compositional reasoning.

The sequence is not a literal phylogenetic ladder.

Honeybees can learn and transfer same/different relational rules to novel stimuli (Giurfa et al., 2001), supporting the narrower claim that transferable relational discrimination need not require a human-sized brain.

Cuttlefish have shown what-where-when behavioral organization meeting standard criteria for episodic-like memory (Jozet-Alves, Bertin, & Clayton, 2013), and later work demonstrated substantial delayed control in a delay-of-gratification task (Schnell et al., 2021).

These examples support the narrower conclusion that component capacities relevant to human cognition occur in distant lineages and can be realized by substantially different nervous-system architectures. They do not establish a single linear ancestry of cognition, nor do they demonstrate that nonhuman animals execute human formal logic or possess phenomenology identical to human cognition.

8.8 Physical witness and semantic inflation

A danger in comparative cognition is to redescribe simple behavior with sophisticated human concepts.

UCT therefore introduces a Physical Witness Criterion:

A claimed higher-order relation belongs to the actual constitutive organization only if the physical process itself distinguishes that relational configuration under controlled, physically admissible perturbation.

For example, “same/different” should not be attributed merely because an analyst can describe stimuli relationally. Transfer to novel physical stimuli provides stronger evidence that the system is sensitive to an invariant relation, though behavioral transfer alone still need not reveal the complete ontic mechanism.

The criterion blocks:

observer’s description $\Rightarrow$ invented phenomenal dimension.

8.9 Body-centered organization

With increasingly coordinated sensorimotor organization, a system can physically encode relations involving:

- internal/external boundary;
- body position;
- sensor position;
- action state;
- consequences of self-generated action;
- persistent internal variables.

Write this only as a projection of already existing process organization:

$$\mathcal{S}(M) = \Pi_{sys}[\mathbf{D}^{ontic}(M)].$$

This is system-relative organization, not a new observer inside the system.

It can support increasingly rich body-centered and agency-related experience without being the origin of experience itself.

8.10 Self-model

A self-model is a high-level physical organization in which some internal states track, predict, compress, or regulate aspects of the system's own organization across time.

Possible ingredients include:

- body state;
- action consequences;
- memory;
- prediction;
- social information;
- control;
- metacognitive access.

UCT does not define the self-model as the subject, and it does not make self-modeling necessary for basal experience.

8.11 The conceptual "I"

Human conceptual selfhood adds language, autobiographical structure, explicit temporal narration, symbolic categorization, social cognition, metacognition, and culturally transmitted conceptual tools.

A system can then instantiate structures corresponding to propositions such as:

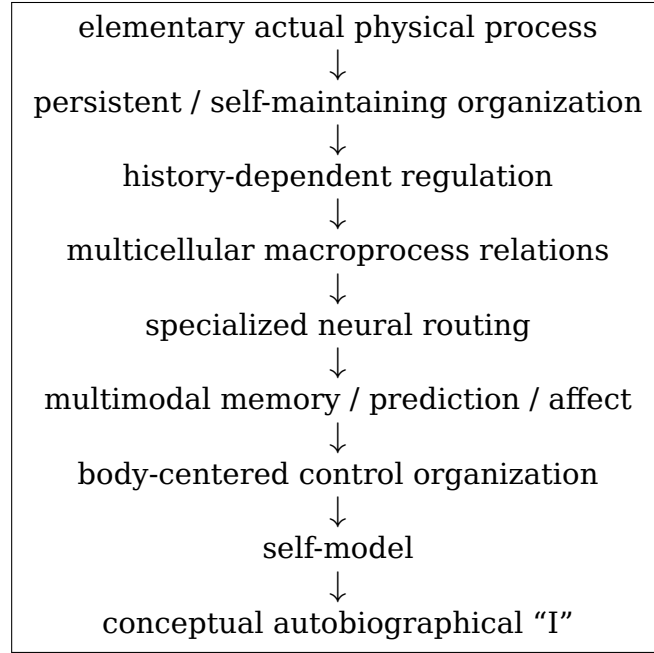
I am the one undergoing this experience.

or:

I remember being the same individual who experienced that event.

Within UCT, this conceptual "I" is a late, high-dimensional organization inside experience, not the entity that creates experience.

The proposed structural sequence is therefore:



The arrows mean structural transformation, not one linear species ancestry.

The comparative cases in this section are therefore evidential anchors for organizational possibilities and intermediate capacities. They are not empirical confirmation of universal experience, structural identity, or cross-species quale identity.

8.12 No universal consciousness ladder

Evolution can:

- add relations;
- delete relations;
- reweight couplings;
- prune structure;
- specialize;
- compress irrelevant details;
- introduce new temporal dependencies;
- bifurcate dynamically;
- lose capacities;
- converge on similar functions by different architectures.

Therefore UCT rejects:

$$\frac{dC}{dt} > 0$$

for a universal scalar amount of consciousness.

The more general representation is:

$$\lambda \mapsto \mathbf{D}_{\lambda}^{ontic}(P_{\lambda}) \mapsto \Phi_{\lambda}(P_{\lambda}).$$

Evolution is a path through a branching space of experiential organization, not a staircase from unconscious matter to conscious humans.

9. Structural transformation versus enrichment

A previous development branch emphasized relational enrichment:

$$D_1 \prec_E D_2.$$

Strict enrichment remains useful when D_2 preserves the distinctions of D_1 while adding new, nonredundant constitutive relations.

But biological change is more general.

Let:

$$T : \mathbf{D}_1 \mapsto \mathbf{D}_2.$$

Possible transformations include:

- current-state change;
- coupling reweighting;
- relation addition;
- relation deletion;
- pruning;
- factorization or refactorization;
- temporal-depth change;
- abstraction/compression;
- dynamic bifurcation;
- specialization.

Thus:

$\text{enrichment} \subset \text{structural transformation}.$

Under A5-OI:

$$\mathbf{D}_1 \not\cong \mathbf{D}_2 \Rightarrow \Phi_1 \not\cong \Phi_2.$$

This is a direct consequence of the identity commitment, not an independent discovery or an additional empirical law.

9.1 An exact edge-addition family with unchanged declared output

A concrete consequence should identify which relation changes, rather than merely restating that different complete structures have different experiential types. Extend Appendix B by controlling the two cross-links separately:

$$s^+ = s \oplus u, \quad a^+ = s \oplus (\alpha b), \quad b^+ = s \oplus (\beta a), \quad y = s, \quad (\alpha, \beta) \in \{0, 1\}^2.$$

The fixed comparison contains the whole network $W_{\alpha\beta}$, the implemented s -process, and the boundary-conditioned $a \mid b$ constituent split. Buffers, the clock, state-reset ports, and idealized completeness are as in Appendix B. The settings are distinct mechanisms, not one continuous A2 interpolation. Conditional UCT statements concern their actual realizations, not uninstantiated equations.

For common s_0 and input string, all four settings have the same declared output. To measure retention of an *initial internal perturbation*, fix the common boundary trajectory s_t , allow all four initial pair states $z_0 = (a_0, b_0)^T$, and apply no further

pair resets. Differences between two pair trajectories obey linear dynamics over $\mathbb{F}_2$:

$$\delta z_n = A_{\alpha\beta}^n \delta z_0, \quad A_{\alpha\beta} = \begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix}.$$

The identical boundary drive cancels from the difference equation. Define the retained distinguishability at horizon $n \geq 1$ by the number of different pair states reachable from the four initial pair states under that fixed drive:

$$N_n = |\{z_n : z_0 \in \mathbb{F}_2^2\}| = 2^{\text{rank}(A_{\alpha\beta}^n)}.$$

This is a specific response-partition invariant. It is not the number of experiences, a scalar amount of consciousness, a familiar quale label, or a full phenomenal metric.

9.2 Proposition: three distinct causal-retention regimes

Within this declared family, the following statements hold for every $n \geq 1$:

Cross-links	N_1	$N_n, n \geq 2$	Directed pair cycle	Whole-update image size
(0, 0)	1	1	absent	2
(1, 0)	2	1	absent	4
(0, 1)	2	1	absent	4
(1, 1)	4	4	present	8

Proof. For (0, 0), the pair matrix is zero. With exactly one cross-link it has rank one and square zero. With both cross-links it is the swap matrix and has square equal to the identity; all its powers are invertible. Rank-nullity gives the values of N_n . At fixed input u , the whole transition first determines $s^+ = s \oplus u$, so the images from the two different current s values are disjoint. For each such value the pair map has $2^{\text{rank } A}$ images, giving $2 \cdot 2^{\text{rank } A} = 2, 4, 4, 8$. The pair has a directed two-edge cycle precisely when both cross-links are present. $\square$

Consequently, adding a first link creates one-step distinguishability of some initial pair perturbations; adding the complementary link creates indefinitely sustained distinguishability in this exact noiseless model. Neither single link alone produces that sustained response property. With one link, the fixed split already fails independent component-update factorization despite lacking a directed pair cycle. Nonproduct organization, recurrence, and persistence are therefore different properties, not interchangeable labels.

The all-horizon result follows from $A^2 = 0$ or $A^2 = I$, not from extrapolating a finite simulation. An independently executable check enumerates the finite transition and intervention cases and checks representative horizons as a regression test. In non-ideal or noisy systems indefinite retention is not implied by a finite feedback diagram; the noise model and stability would have to be analyzed.

9.3 The conditional experiential interpretation and its limit

Image cardinality is preserved under bijective state conjugacy. The settings with image sizes 2, 4, and 8 therefore have different complete **modeled** transition types. If these are actual tokens and the stipulated organization is complete for them, A5-OI entails different whole-token experiential types. A5-SI says more specifically that

their distinctions, causal relations, and temporal-retention relations belong to the same tokenwise organization in experiential presentation. The result does not assert that the bit count $\log_2 N_n$ measures how much experience exists.

Throughout the family the implemented s -process has the same current state, update, and represented interfaces under common s_0, u . A5-W consequently gives its type equivalence under the same actuality/completeness qualifications. These sub-process and whole-process comparisons are fixed in advance. They are not alternative boundaries selected to rescue an unwanted result.

The one-link settings are related by exchange of a, b when the comparison permits that physical port permutation and maps the pointed state accordingly. With fixed individually named intervention roles, their directed roles remain distinguishable. This makes the covariance convention explicit rather than choosing it after the conclusion.

9.4 A shared physical example, not an independent proof of identity

The mathematics uses specified dynamics, boundary conventions, and admissible resets. It does not require A1 or A5 to calculate N_n . UCT supplies a conditional experiential interpretation of independently specified organizational facts; the numbers do not independently verify that interpretation. An output-only whole-token equivalence hypothesis identifies all four settings, while UCT's complete-organization hypothesis separates the three image-size classes under its stated premises. This is a difference of conditional theoretical commitments, not an observed phenomenal discrimination or a refutation of every functionalist account.

UCT II v1.1 uses the same family to separate causal retention, graph recurrence, and unchanged declared consumer output, while retaining target-specific semantic and biological bridges. The point is a constrained connection between the two papers: a precise relation is added, a calculable temporal distinction changes, and neither a new existence gate nor a new experience substance is introduced. Nilpotence, invertibility, and response partitions are established mathematical tools; no priority claim is made for that mathematics or for the general possibility of behaviorally silent internal change.

10. Scientific bridge architecture

A5-OI concerns complete ontic organization and complete experiential structure.

Ordinary science observes neither completely.

Finite empirical tests therefore require explicit auxiliary structures.

10.1 Declared phenomenal target

Let:

$$\Psi_v(P)$$

be a finite, preregistered phenomenal/psychophysical target inferred from some combination of:

- similarity judgments;
- discrimination performance;
- subjective ratings;

- confidence;
- no-report perceptual indicators;
- multidimensional report patterns;
- physiological measures with independently justified interpretation.

By default:

$$\boxed{\Psi_v(P) \neq \Phi(P).}$$

It is a scientific projection, not complete phenomenology.

10.2 Sufficiency and reflection are different bridge hypotheses

Let D_v be a frozen finite physical view and Ψ_v a declared finite phenomenal target. A **sufficiency bridge** $B_v^{\rightarrow}$ asserts

$$D_v(A) \cong D_v(B) \implies \Psi_v(A) \cong \Psi_v(B).$$

Equivalently, on the compared classes, the target factors through a map f_v from the physical view. This does not require that f_v be injective. To assert a biconditional one must separately impose a **reflection bridge** $B_v^{\leftarrow}$:

$$\Psi_v(A) \cong \Psi_v(B) \implies D_v(A) \cong D_v(B).$$

Together these bridges say that the physical view and the finite target distinguish the same classes within the declared scope. A5-OI for complete structures does not imply reflection after either side has undergone a lossy projection. For example, even an identity correspondence between two full physical and phenomenal states can have an observed target that assigns both states the same response. The finite target then remains sufficient as a function of the physical state but fails reflection.

10.3 Operational Structural Correspondence

Two packages should therefore be distinguished:

$$\begin{aligned} OSC_v^{\rightarrow} &= A5 + B_v^{\rightarrow} + \text{measurement assumptions,} \\ OSC_v^{\leftrightarrow} &= A5 + B_v^{\rightarrow} + B_v^{\leftarrow} + \text{measurement assumptions.} \end{aligned}$$

The notation denotes packages of assumptions, not numeric addition. The operational implications are supplied by the bridge conditions; attaching A5 explains their theoretical interpretation and does not constitute a separate empirical derivation of those conditions. The one-way package predicts target equivalence for equivalent frozen physical views. The two-way package additionally predicts target non-equivalence for non-equivalent views. Reflection must be defended for the specific target; it cannot be supplied merely by requesting more confidence in the same coarse report.

Approximate applications must declare physical and phenomenal distance functions, equivalence tolerances, and error models. Failure to reject a measured difference is not evidence of equivalence without a suitable equivalence-testing design and adequate precision. The exact finite examples below do not solve this measurement problem.

A finite empirical failure challenges the operational package. It does not automatically isolate A5 from either bridge direction or from measurement assumptions.

11. Physical Token/View Selection Protocol

The main falsifiability danger for UCT is post-hoc redefinition.

If a failed experiment can always be answered by choosing a new token, scale, horizon, intervention family, or structural descriptor after seeing the phenomenal result, the theory becomes scientifically non-vulnerable.

Version 1.1 therefore proposes the Physical Token/View Selection Protocol (PTVSP).

Before phenomenal unblinding:

1. declare candidate actual-token hypotheses;
2. declare admissible physical views;
3. declare physical intervention families;
4. declare held-out physical response targets;
5. evaluate physical predictive sufficiency;
6. evaluate stability under measurement perturbation;
7. evaluate cross-system reproducibility when possible;
8. prefer parsimonious physically sufficient views or a preregistered Pareto set;
9. freeze token/view/bridge definitions;
10. only then join or reveal phenomenal outcomes.

The central rule is:

No Post-Hoc Token/View Rescue.

A failed preregistered bridge remains failed.

Changing token, view, grain, horizon, or bridge afterwards creates a new version of the hypothesis.

12. Empirical failure conditions

12.1 Same frozen physical view, different phenomenal target

If

$$D_v(A) \cong D_v(B), \quad \Psi_v(A) \not\cong \Psi_v(B),$$

then the frozen one-way operational bridge fails, subject to its error model. It would be circular to retain that very implication as an independently secure premise while using its counterexample to isolate A5. The failure directly concerns the view/target bridge, not automatically the full identity axiom.

A countermodel explains the distinction: take two different complete physical structures, identify each tokenwise with its experiential structure, let the finite physical view map both to one value, and let the finite phenomenal target distinguish them. A5-SI is satisfied, while the proposed finite sufficiency bridge fails. Thus finite physical underdescription alone can produce this pattern.

A challenge specifically to A5-W would instead require independently supported equivalence of the **complete constitutive** physical organizations and a valid phenomenal observation that cannot differ between isomorphic full experiential structures. Those are stronger premises than agreement on a frozen coarse view. They

must not be smuggled in by renaming the failed finite prediction “completeness.” In ordinary experiments the operational package is the direct test target.

12.2 Different frozen physical views, identical phenomenal target

Under a declared two-way operational bridge, the pattern

$$D_v(A) \not\cong D_v(B), \quad \Psi_v(A) \cong \Psi_v(B)$$

refutes the finite reflection direction. The same full-identity construction with an information-preserving physical view and a constant phenomenal readout satisfies A5-SI while failing reflection. A difference can therefore be discarded by measurement even when it belongs to the full structure.

To challenge the converse full-structure implication, one would need independently supported complete physical nonisomorphism **and** complete experiential equivalence, not merely equivalent reports. This paper does not claim that routine measurements supply those premises.

A5 is not needed to derive operational predictions that have already been stipulated in a bridge. A successful bridge test supports that fixed bridge/model against specified alternatives; it is not, solely for that reason, differential evidence for A5 over every other interpretation using the same operational law. Any stronger model comparison must expose independently motivated constraints that generate different likelihoods or predictions. This is a statement about empirical discrimination, not an additional foundational mechanism of experience.

12.3 Behaviorally matched causal rewiring

A particularly informative design would match:

- overt behavior;
- report;
- task accuracy;
- confidence;
- access variables;

while manipulating a preregistered constitutive causal relation that survives the frozen physical view.

For actual tokens whose complete constitutive structures differ, A5-OI predicts a difference in complete experiential organization even when the chosen output variables match. A finite measurement difference additionally requires the corresponding reflection bridge. If every available phenomenal indicator has been matched by design, the remaining structural claim has not thereby become an observed empirical discrimination.

This places UCT within the empirical pressure created by unfolding/substitution arguments against causal-structure theories (Doerig et al., 2019; Kleiner, 2020; Tsuchiya et al., 2020; Usher, 2021; Herzog et al., 2022; Basri, BarAvi, & Hemmo, 2026).

12.4 Latent-memory test

Match explicit current report and task content while preserving a preregistered difference in latent constitutive memory dispositions that changes future admissible response.

A5-OI predicts a difference in complete experiential organization, though not necessarily a difference in explicitly recalled content.

This is a conditional consequence of the injective identity commitment; other sufficiently strong causal-structural accounts may share it. A distinguishing experiment needs a named competing prediction and a separately justified sensitive phenomenal target. Internal-model verification alone does not supply either.

13. Relation to UCT II and neighboring theories

UCT II asks whether operational cores of major consciousness theories can be re-constructed, embedded, reinterpreted, or shown to conflict as effective organization theories on top of the UCT mother formalism.

The v1.1 interface is explicitly view-based:

$$D_v^{PA}(P) := \Pi_v[\mathbf{D}^{ontic}(P)], \quad O_T = \mathcal{T}_T(D_v^{PA}(P), B_T).$$

An estimate is denoted $\widehat{D}_v^{PA}$, and an underidentified view yields an output set rather than an arbitrarily selected point. The tuple formerly written $\mathfrak{D}^{PA}$ in UCT II v1.0 is a physically anchored scientific representation when used in finite translation, not a license to identify every chosen view with complete ontic organization. Reproducing target-theory computations requires its fixed bridge conventions, not A1. No-universal-gate relocation additionally uses A1 and a scoped nonuniversality witness; interpreting physical organization as experiential uses the relevant A5 commitment. Finite phenomenal prediction needs a separate operational bridge. The shared example in Section 9 respects all four distinctions.

The relation is asymmetric:

$$\boxed{\text{failure of one UCT-II bridge} \not\Rightarrow \text{failure of the basal ontology}}$$

but:

$$\boxed{\text{failure of A1/A5} \Rightarrow \text{revision of the UCT-specific interpretation of UCT II.}}$$

UCT II introduced explicit Bridge Admissibility Conditions, Translation Fidelity Requirements, formal recovery classes, and separate empirical-validation levels. Those constraints remain essential for preventing a rich mother formalism from trivially “containing” other theories by lookup.

Version 1.1 does not claim novelty for:

- panexperientialism or panpsychism;
- process-oriented metaphysics of experience;
- mind-brain or mind-physical identity strategies in general;
- causal/phenomenal structuralism;
- multiscale process accounts;
- overlapping conscious systems;
- mechanistic physical implementation criteria;
- structural approaches to subjectivity;
- the general idea that theories may be partially complementary.

Several neighboring traditions overlap with individual UCT commitments. Panexperientialist and process traditions already treat experience as widespread or fundamental rather than as a late biological product (e.g., Rosenberg, 2004, building on earlier process traditions). IIT explicitly advances an explanatory identity between phenomenal structure and a substrate’s intrinsic cause–effect structure (Albantakis et al., 2023), while neurophenomenal structuralism develops a relational/structural account of specific phenomenal character (Lyre, 2022). General mind–brain identity theories likewise deny that mental and physical processes are merely correlated. UCT therefore does not claim to originate the idea of identity itself.

The combination, de-combination, overlap, and individuation problems are also established literatures (Chalmers, 2016; Miller, 2018; Blackmon, 2021). Recent preprint-level structural-subjectivity programs go further by proposing positive structural criteria for the presence or locus of subjectivity; for example, Nagae (2026) identifies subjectivity with viability-constrained irreversible exclusive commitment. UCT differs at this point: it does not use action, viability, irreversibility, recurrence, access, self-modeling, or any other nonuniversal organization as a basal experience-existence gate, and it presently offers only a scoped no-free-selection theorem rather than a universal positive subject criterion.

Neighboring and critical literatures also include Information Closure Theory (Chang et al., 2020), universal-theory criteria (Kanai & Fujisawa, 2024), mechanistic computation and implementation accounts (Piccinini, 2015; Anderson & Piccinini, 2024), and the unfolding/substitution debate (Doerig et al., 2019; Kleiner, 2020; Tsuchiya et al., 2020; Usher, 2021; Herzog et al., 2022; Basri et al., 2026).

The contribution claimed here is therefore not any single ingredient. This revision explicitly formalizes the intended tokenwise identity and distinguishes it from its weaker pairwise consequence. Product-isomorphism reasoning, quotient closure, and intervention consistency use established mathematics. The claimed contribution is the following constrained conjunction and dependency architecture:

universal process-token experience
 +ontic-token structural identity
 +nested non-exclusion
 +token/type/lineage separation
 +composition under tokenwise structural identity
 +experience/subject separation
 +restricted selection obstruction in an overlap orbit
 +explicit ontic/view/estimate firewall
 +failure-prone finite bridge methodology.

The earlier targeted prior-art audit did not locate an exact match for the rc3 package. That observation neither certifies historical priority nor establishes that a conjunction of familiar elements is a major advance. Rc4 adds proof repairs and a finite worked instance; it does not claim priority for the orbit argument, product-preservation mathematics, causal abstraction, or the general possibility of behaviorally silent internal change. Its scientific value must be assessed through the explanatory restrictions of the package and future independently constrained applications.

14. Strong objections and limitations

14.1 The axioms are strong

A1 and A5-SI are not empirical discoveries.

They are strong starting commitments.

The formal theorems derived from them must therefore be evaluated partly by the explanatory costs and empirical obligations of the axioms.

14.2 A5-OI risks tautological explanatory rhetoric

Once structural identity is assumed, results of the form:

$$\mathbf{D} \not\cong \mathbf{D}' \Rightarrow \Phi \not\cong \Phi'$$

are straightforward consequences.

They should not be marketed as deep independent discoveries.

The nontrivial content must come from:

- process individuation;
- physical anchoring;
- exact no-go consequences;
- multiscale relations;
- subject non-selection;
- independently constrained bridge predictions.

14.3 Complete ontic organization is not directly observed

Science works with:

$$D_v$$

and:

$$\widehat{D}_v.$$

Direct empirical isolation of full A5-OI from bridge assumptions is therefore difficult. UCT should not disguise this epistemic limitation.

14.4 Complete phenomenology is not directly measured

Human report, behavior, psychophysics, and physiology are projections.

Even convergent measures do not automatically equal complete phenomenal structure.

This limits the strength of empirical falsification claims.

14.5 Process proliferation

A permissive process ontology can yield many weak, overlapping, or nested experience-bearing tokens.

UCT accepts such plurality as part of its intended ontology. The number of tokens is not a problem to be eliminated by an arbitrary experience threshold; the scientific task is to distinguish their organizations and relevance to a specified target.

Scientific salience and experience existence are separate questions.

14.6 Macroprocess descriptions are applications, not new existence gates

UCT accepts actual causal processes at different scales. The finite criterion in Section 2.4 makes that domain concrete in one classical representation. No unique natural macro boundary, autonomous coarse state, or maximal integration score is required before an actual process can possess experience. In concrete applications, competing claims about physical boundaries and constitutive relations remain accountable to the system being described. Uncertainty in that physical identification is not evidence that the framework lacks a separate macro-experience mechanism.

14.7 Subject terminology must not become a prerequisite

If subject names an experience-bearing process, the ontology supplies many possible referents. If it names a conceptual owner, self-model, or one exclusive human-like agent, additional organizational descriptions are relevant to that usage. UCT need not add an extra owner of experience to explain why processes have experience. The optional selection theorem constrains exclusive labeling rules; failure to specify one universal subject partition is not a foundational incompleteness criterion adopted by this paper.

14.8 General quality identity and biological specification

A5-SI supplies the same general account of experiential quality for elementary processes, cells, animals, and humans: the quality is the intrinsic presentation of the actual organization. Human complexity does not introduce a second explanatory gap inside the theory. A1-A6 do not provide a biological lookup table assigning familiar labels such as red or unpleasant to every measured neural configuration. Obtaining such mappings is a task for particular scientific applications, just as identifying a concrete cellular organization is. It must not be misreported as absence of the general physical-experiential principle.

14.9 Cross-species qualia identity is underidentified

Similar behavior, homologous mechanisms, or structurally comparable organization do not by themselves prove identical subjective qualities across species.

Claims should therefore use terms such as:

- precursor;
- conserved mechanism;
- structural homology;
- convergent similarity;

rather than asserting uncalibrated qualitative identity.

14.10 Relativistic and quantum extensions

The present finite/classical formalism is not a final fundamental-physics theory.

A relativistic version should be formulated over causal spacetime histories rather than a preferred global time slice.

A quantum version requires appropriate state spaces, channels/instruments, subsystem criteria, and intervention notions.

14.11 Semantic representation

Causal tracking does not automatically establish semantic aboutness.

Detailed models of semantic representation, conceptual content, and explicit self-reference belong to higher-order applications. They do not supply prerequisites for basal experience or a second physical-experiential principle.

14.12 Formal strength and finite measurements must remain explicit

A5-SI is a strong interpretation of identity. Its tokenwise isomorphisms support the product proof; the weaker A5-OI formula by itself does not. Rc4 records this difference rather than attributing the stronger formal statement retroactively to the old formula. A critic may reject the identity commitment; that is distinct from an internal derivation failing to follow from it.

Finite scientific measurements can discard distinctions even when complete experiential and physical organizations share the same structure. Therefore finite reflection remains an additional condition of an operational two-way test, not another experience-generating mechanism or another foundational axiom. The measurement qualifications do not replace the unified explanation of experience.

15. Claim-status summary

Claim	Status in v1.1
Universal experience for actual valid process tokens	Axiom A1
Scoped continuity of experiential organization	Axiom-schema A2
Embedding does not terminate a continuing token	Axiom A3
Human-like conceptual self not required for basal experience	A4 boundary commitment
One-way phenomenal completeness	inherited A5-W
Tokenwise structure-for-structure identity	A5-SI: explicit formalization of the intended identity; stronger than the rc3 displayed formula
Pairwise type equivalence	A5-OI consequence of A5-SI
Token/type/lineage distinction	Axiom A6
Organization-Gate Impossibility	theorem conditional on A1 + nonuniversality witness
No Hidden Quale	consequence of A5-W
Probe Invariance	consequence of ontic/probe separation + A5-W

Claim	Status in v1.1
No Future Contamination	consequence of A5-W
Causally Screened History	consequence of A5-W + A6
Non-Summative Combination	actual tokens + A5-SI + standard product in the declared common category
Extra experience-generating composition mechanism	not introduced
A5-OI alone transports independent products	disproved as an implication of the displayed formula
Macro invariance with micro change	macro equivalence: A5-W; lower-type difference: A5-OI
Symmetric-overlap non-selection	orbit lemma plus explicit nonempty/exclusive candidate restriction
Intrinsic structure rules out every invariant partition	not established; explicit counterexamples
Canonical positive subject selector	not a required universal element of UCT
General account of human/cellular quality	one A5-SI identity principle
Familiar quality labels mapped to measured mechanisms	application-level biological/psychophysical task
Structural evolution from physical process to conceptual self	explanatory/application synthesis
Evolutionary continuity entails A1/A5	explicitly rejected
PTVSP	methodological proposal
OSC _v	one-way and two-way packages distinguished
Finite bridge reflection	additional requirement for the two-way package
Exact quotient closure	established mathematical construction, not a tokenhood theorem
Three-register example	exact finite thought-experiment verification; no empirical consciousness validation
Four-setting causal-retention result	stated finite dynamics; algebraic all-horizon proof; conditional A5 interpretation
UCT-II v1.1 interface	scientific view/estimate separated from ontic structure
A1/A5 empirically established	not established
Hard problem deductively solved	not claimed
Extra universal subject owner or partition	not postulated as a prerequisite of experience
Individual ingredients are unprecedented	explicitly not claimed
Exact rc3 conjunction found in prior targeted search	earlier audit located no exact match; does not certify rc4 priority
Historical priority	not established

16. Conclusion

UCT I v1.1 uses one explanatory principle from elementary physical processes through cells, animals, and humans: each actual process has experience, and its experiential organization is the intrinsic presentation of its physical organization. Evolution changes that organization, its discriminative repertoire, temporal relations, and higher-order self-representations. It does not require a new experiential substance when a cell becomes part of a multicellular process or when human conceptual selfhood appears.

This edition formalizes this intended identity as A5-SI. The tokenwise structural isomorphism is stronger than the earlier displayed pairwise equivalence formula. With the standard product in a common declared structural category, it supports the repaired non-summative combination proof. The distinction between strong identity and weak type correspondence is an explicit formal clarification, not an additional mechanism for generating experience.

The theory accepts nested and overlapping actual processes without requiring one exclusive macro boundary or subject partition. Selfhood and familiar quality labels concern particular organizations and their scientific descriptions; they are not new existence gates. A scoped symmetry theorem explains a limitation of optional exclusive selectors without making the absence of such a selector a defect in the basal ontology.

The exact three-register example illustrates a substantive consequence of token-relative organization: a whole-process structure can change while an implemented sub-process and the declared output remain invariant. The finite calculations verify those model properties. They do not supply independent empirical confirmation of structural identity or a measurement of circuit experience.

The four-setting extension proves exactly which initial perturbation distinctions are erased, retained for one step, or sustained by complementary coupling; these are finite-model organizational results, not scalar consciousness levels. Further theoretical work should extend such controlled transformation analyses and establish faithful links to neighboring theories. Biological and psychophysical work can identify concrete realizations when a particular application requires them. Neither task calls for a second foundational explanation of why human rather than cellular experience exists. The general principle, its conditional deductions, and its empirical status are stated separately. The paired UCT II v1.1 uses a compatible scientific-view interface and preserves both exact mechanistic results and their evidential limits.

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Reference and version note. Earlier editions and internal candidates are retained in the revision record. The three causal-abstraction references introduced during rc4 were checked against the original publisher or author-hosted records; their use does not imply empirical support for UCT. This edition identifies its companion by the new version-specific DOI. Sources support the particular biological or mathematical claims for which they are cited, not A1 or A5 by themselves.

Reference-status note. Literature added specifically for the v1.1 evolutionary/comparative synthesis was checked against publisher, PubMed, or journal records during the 29 September 2026 claim audit. Basri et al. (2026) was indexed by PubMed as ahead-of-print on 8 September 2026 even though the journal issue is dated November 2026. The prior-art audit also added Rosenberg (2004) as an explicit panexperientialist comparison and Nagae (2026) as a recent preprint-level structural-subjectivity comparator.

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Appendix A. Main changes relative to v1.0

1. Replaces the potentially ambiguous scale-indexed ontological reading with a token-relative ontic organization:

$$\Phi(P) \leftrightarrow \mathbf{D}^{ontic}(P).$$

2. Separates:

$$\mathbf{D}^{ontic} \neq D_v \neq \widehat{D}_v \neq Subject.$$

3. Separates the intended tokenwise identity A5-SI from its pairwise A5-OI and one-way A5-W consequences.
4. Marks all converse/nonisomorphism theorems as A5-OI-dependent.
5. Repairs Non-Summative Combination using explicit tokenwise identity A5-SI and the standard product in the declared common structural category.
6. Adds Probe Invariance, No Future Contamination, and Causally Screened History.
7. Adds the restricted candidate-orbit obstruction and explicitly retains possible invariant global partitions.
8. Recasts A3 primarily as process persistence under embedding.
9. Recasts A4 as Selfhood Non-Prerequisite and removes positive self-formation mechanisms from the axiomatic layer.
10. Demotes First-Person Reference Field, binary macrofield membership, H-V*, and RREP-as-universal-generator from the foundational layer.
11. Adds the structural-evolutionary synthesis from elementary physical processes to conceptual selfhood.
12. Introduces the distinction (with one-way sufficiency and two-way reflection separated in rc4):

$$A5 \neq B_v \neq OSC_v,$$

making finite empirical bridge assumptions explicit.

13. Replaces grain-only falsifiability discipline with Physical Token/View Selection Protocol and No Post-Hoc Token/View Rescue.
14. States explicitly that finite experiments ordinarily test an operational bridge package rather than isolated full A5-OI.
15. Distinguishes internal theoretical obligations from application-level physical identification and measurement. General macro-experience gates, a universal exclusive subject selector, and a second human-quality mechanism are not added as prerequisites.
16. Adds the pointed, port-aware finite comparison signature, four-setting causal-retention proposition, noncircular bridge-failure countermodels, and a typed interface to UCT II v1.1.

Appendix B. An exact three-register thought experiment

B.1 Physical specification and scope

Stipulate three physically implemented binary registers s, a, b , a common clock, an external input u , XOR gates, buffered one-way connections from s , and switchable cross-connections between a and b . The switch setting is $\lambda \in \{0, 1\}$. One clock step is

$$\begin{aligned} s_{t+1} &= s_t \oplus u_t, \\ a_{t+1} &= s_t \oplus (\lambda b_t), \\ b_{t+1} &= s_t \oplus (\lambda a_t), \\ y_t &= s_t. \end{aligned}$$

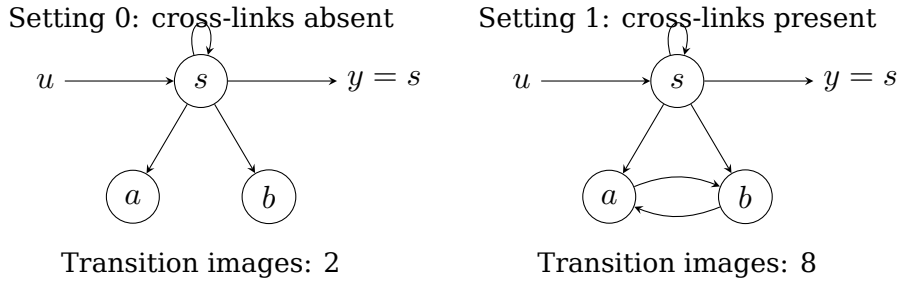


Figure 1: The two stipulated circuit settings. Arrows denote state dependencies. The added pair cross-links distinguish the settings; the external output is the same implemented state in both. Clock and buffering are specified in the text.

Multiplication and XOR act on bits. Buffering is stipulated to exclude feedback and loading from the downstream pair into the s update. In real hardware this isolation and all relevant boundary effects would require examination. The mathematics treats the finite update/intervention specification as complete within the stated idealized model. A real circuit's complete ontic structure is not established by this calculation. No circuit was built and no phenomenal observations were collected.

Before calculating any output, specify two token hypotheses: the whole implemented network W_λ , and the continuing implemented register process S_λ with state s , its update mechanism, clock, and declared boundary u . The whole is causally connected through the outgoing s links in both settings. The cross-coupled a, b pair has boundary s . These are explicit component/boundary choices, not subject assignments. A1 applies conditionally if the stipulated processes are actually instantiated and meet the token requirements; a mathematical state table is not itself proof of that instantiation.

The intervention family comprises all 27 coordinate clamps immediately before an update: each register is either unchanged, set to 0, or set to 1. The boundary input takes both bit values. The switch settings are separately compared mechanisms; they are not assumed to constitute a continuous A2 path. This is a worked theoretical specification, not a preregistered empirical study.

B.2 Exact closure of an implemented sub-process

For $\pi(s, a, b) = s$, the macro update is $\bar{T}_u(s) = s \oplus u$. Every two physical states with the same s have the same next projected state. Projected coordinate clamps act as the same clamp on s if it is targeted and as the identity otherwise. Hence

$$\pi(T_u^\lambda(Ix)) = \bar{T}_u(\omega(I)\pi(x))$$

for every declared clamp, input, setting, and state. The implementation of s is part of the physical specification, not inferred from closure alone. The present state and future update of this sub-process do not depend on a or b , even though it causally drives them. The case separates physical downstream connection from participation in this particular process's state and update.

By induction on time, for the same s_0 and any input string,

$$y_t = s_0 \oplus u_0 \oplus \cdots \oplus u_{t-1} \quad (t \geq 1)$$

in both settings, independently of initial a, b . This is equality of the declared output behavior, not equivalence under every conceivable additional physical experiment.

B.3 Full structural difference is invariant to state relabeling

Fix either input value. At $\lambda = 0$, both downstream next states equal s , so the eight-state transition map has exactly two images. At $\lambda = 1$, the map is bijective: from (s', a', b') , recover

$$s = s' \oplus u, \quad a = b' \oplus s, \quad b = a' \oplus s.$$

It therefore has eight images. Conjugating a transition map by a bijective state relabeling preserves its image cardinality. Consequently the two whole-network transition structures are not isomorphic, even before requiring isomorphisms to preserve the intervention ports.

At a fixed boundary s , the $a \mid b$ split has independent component updates at $\lambda = 0$. At $\lambda = 1$, changing b changes a^+ and changing a changes b^+ ; no product of component-local update maps reproduces this behavior relative to these physical ports. This is a statement about the specified split, not about every conceivable factorization or about one phenomenal subject. Because the boundary is explicitly conditioned, correlation from the common s drive is not mistaken for cross-coupling.

The two distinctions yield compatible conditional UCT statements:

Fixed comparison	Mathematical result	Conditional UCT interpretation
Whole tokens W_0, W_1	Nonisomorphic full modeled dynamics	A5-OI gives different complete experiential types if the stipulated organization is complete and the tokens actual
Implemented processes S_0, S_1	Same state/update and represented interfaces	A5-W gives equivalent experiential types under the same completeness and actuality qualifications

Fixed comparison	Mathematical result	Conditional UCT interpretation
Output y	Same trajectory for common s_0, u	Output equality does not settle whole-token experiential equivalence
Downstream split $a \mid b$, boundary s fixed	Cross-dependence appears at setting 1	A compositional interpretation uses A5-SI and the specified constituent product; it does not determine one subject

The coexistence of these conclusions is the point of token-relative completeness. Replacing a whole-process claim by the sub-process claim after a failed test would still violate the freeze protocol. Here both comparisons are specified in advance and are reported together.

B.4 Two checks against arbitrary coarse-graining

The feature $q = a \oplus b$ obeys $q^+ = \lambda q$. Its exact algebraic closure does not establish a separate actual q -token. No separate parity register is stipulated. A distributed realization could still be proposed and independently examined; absence of a literal register is not a proof of absence of such a process. This example demonstrates a limit of the certificate: algebra alone does not finish the realization argument.

The feature $r = s \oplus a \oplus b$ fails even the deterministic closure test at $\lambda = 0$. With $u = 0$, states $(0, 0, 0)$ and $(1, 1, 0)$ both have $r = 0$, but their next r values are respectively 0 and 1. Thus no deterministic one-state update for r alone exists in that scope. This is a model failure, not a no-experience conclusion.

B.5 What the example does and does not distinguish

An output-sufficiency hypothesis assigning whole-token experiential equivalence whenever every declared u -to- y response agrees conflicts with A5-OI on this model, conditional on the stated actuality/completeness premises. This comparator is deliberately narrow. It is not all functionalism: theories that include internal counterfactual organization need not identify the two whole networks. No IIT 4.0 quantity or GNWT neural model is calculated here; this construction is not reported as an empirical victory over either theory.

The example is a finite specialization and audit of the project’s earlier behaviorally silent internal-change argument. Its value is to put output equality, full nonisomorphism, fixed token boundaries, intervention closure, and the limits of coarse descriptions in one independently checkable specification. It does not claim that the general phenomenon or the underlying mathematics is new.

B.6 Exact verification record

The accompanying standard-library Python script `verification/check_rc4.py` checks 13 scoped properties. It enumerates all 32 state/input/setting transitions, 128 same-projection closure comparisons, and 864 state-reset intervention cases. It also verifies the finite-set compositional counterexample, the restricted symmetry orbit and invariant global counterpartitions, the lossy-target counterexample, the

transition-image invariant, and the coarse-projection examples. Full transition data and results are supplied in `transition_table.csv` and `verification_results.json`. The original 13-check script is preserved and rerun in the v1.1 package. A separate `verification/check_transformations.py` checks the four-setting extension, port covariance, and bridge countermodels. General statements rely on the analytic proofs above; finite enumeration is not a proof about arbitrary physical systems and is not empirical validation of any consciousness axiom.

Appendix C. Revision-specific dependency ledger

Result	Required premises	Scope boundary
Pairwise type equivalence	Tokenwise A5-SI in the common signature	Weaker A5-OI alone does not establish A5-SI
Repaired combination	Actual whole/parts; tokenwise A5-SI; standard constituent product	Relative nonproduct structure, not a subject count
Restricted selection obstruction	One symmetry orbit with overlap; equivariance; nonempty exclusive output	Does not exclude other invariant partitions
Actual finite process criterion	Actual events; inherited causal links; connected subhistory; represented boundary	Describes the domain; no new experience threshold
Finite closure lemma	Surjective projection; representative-independent projected next state	Optional scientific model consistency
One-way measured correspondence	Frozen view/target; sufficiency; measurement assumptions	Not necessarily injective after projection
Two-way measured correspondence	One-way package plus finite reflection	Does not isolate the identity axiom from all auxiliaries
Whole-model divergence	Specified actual realization/completeness; image-cardinality invariant; A5-OI	No independently measured phenomenal difference
Sub-process invariance	Same complete implemented sub-process; A5-W	Does not assert whole-network invariance
Human and cellular quality	The same A1 and A5-SI principles	Concrete biological identification is a separate application
Pointed, port-aware isomorphism	Declared signature; fixed point, ports, operations and temporal relations	A finite model is not a certificate of complete ontic organization
Four-setting retention partition	Stated binary updates; held common boundary; initial perturbation; rank-nullity	Counts response distinctions, not consciousness or subjects
Finite bridge failure does not isolate A5	Explicit lossy-view and lossy-target countermodels	No preservation of a falsified implication as a secure premise
UCT-II operational reconstruction	Frozen physical view, shared operators, applicable fixed bridge	Does not alone establish A1/A5 or sensitive phenomenal measurement

The earlier static dependency-map audit checked the presence and direction of encoded dependencies. It did not verify that the displayed weak formula expressed all of the intended strong identity. Rc4 makes that difference explicit. The old freeze candidate should not be reused as a semantic proof certificate without updating the identity and composition nodes.

Appendix D. Optional checks on a finite coarse description

These checks concern a proposed scientific model, not the existence of experience. Actual processes need not admit autonomous, closed coarse descriptions. The finite process criterion in Section 2.4 does not depend on the checks below.

A useful model dossier identifies the actual referent and its physical state carriers or distributed organization; declares the time horizon, boundary, and omitted feedback; and distinguishes implemented relations from a calculated summary. Predictive success alone does not create a new token. A literal register is not required for a distributed realization.

For a deterministic lower-level update $x^+ = T_e(x)$ and a surjective coarse map $\pi : X \rightarrow M$, a closed deterministic update $\bar{T}_e$ exists exactly when

$$\pi(x) = \pi(x') \implies \pi(T_e x) = \pi(T_e x').$$

Proof. If $\pi T_e = \bar{T}_e \pi$, equal coarse states have equal coarse successors. Conversely, define $\bar{T}_e(m) = \pi(T_e x)$ using any representative of m . The displayed condition makes the definition representative-independent, and surjectivity supplies a representative. $\square$

For a declared state-reset intervention I , an additional intervention correspondence ω can be checked by

$$\pi(T_e(Ix)) = \bar{T}_e(\omega(I)\pi(x)).$$

The reset is applied before the transition. Mechanism-changing interventions require separately indexed update maps; the reset equation does not automatically cover them. Approximate, probabilistic, or history-dependent models need their own error and sufficiency conditions. Failure of a closed deterministic description is a failure of that model, not absence of experience in the actual process.

This mathematics belongs to established lumpability and causal-abstraction methodology (Buchholz, 1994; Rubenstein et al., 2017; Beckers & Halpern, 2019). UCT claims no mathematical priority. Appendix B supplies both an implemented closed process and examples illustrating why a closed calculated feature is not automatically another token, and why an arbitrary feature can fail closure. These distinctions keep optional modeling requirements separate from the common explanatory principle for elementary, cellular, and human experience.